

Definition of the Definite Integral

1

Nicholas is approximating the area under a curve using a right-hand sum. The right-hand sum he's written down is

$$\sum_{k=1}^{100} x_k^3 \Delta x \text{ where } \Delta x = \frac{1}{25} \text{ and } x_k = 2 + k\Delta x \quad (1)$$

- (a) Fill in the blanks: Nicholas is approximating the area under the curve $y = \underline{\hspace{2cm}}$ from $x = \underline{\hspace{2cm}}$ to $x = \underline{\hspace{2cm}}$ using a right-hand sum with $\underline{\hspace{2cm}}$ rectangles.
- (b) Use Wolfram Alpha (<http://www.wolframalpha.com/>) to evaluate Nicholas's sum. Note that you'll need to plug the formulas for x_k and Δx in to get something that Wolfram Alpha can evaluate; that is, we can rewrite line (1) as:

$$\begin{aligned} \sum_{k=1}^{100} x_k^3 \Delta x &= \sum_{k=1}^{100} (2 + k\Delta x)^3 \Delta x && \text{(plugging in } x_k = 2 + k\Delta x) \\ &= \sum_{k=1}^{100} \left(2 + \frac{k}{25}\right)^3 \frac{1}{25} && \text{(plugging in } \Delta x = \frac{1}{25}) \end{aligned}$$

and you can tell Wolfram Alpha to evaluate this by typing

`sum (2 + k/25)^3 * 1/25, k from 1 to 100`

- (c) By hand, use a right-hand sum with 2 rectangles to approximate the area Nicholas is interested in.
- (d) Suppose Nicholas wants to use R_n , a right-hand sum with n rectangles (rather than the specific number of rectangles he was using before), to approximate the area he's interested in. Write an expression like that in line (1) that gives R_n . (Your answer should be in terms of n .)
- (e) Use Wolfram Alpha to evaluate your expression for R_n . Please show your work to determine what you enter into Wolfram Alpha (This is basically like (b), except that you're evaluating your answer to (d). Your answer will be in terms of n , since the value of R_n certainly depends on the number of rectangles.)
- (f) Is your answer to (e) consistent with your answer to (c)? Explain briefly.

(g) To find the exact area Nicholas wants, what limit should you take? Evaluate this limit by hand (using your answer to (e)).

(h) **Reflect Back.** In this problem, you used the definition of the definite integral as a limit of Riemann sums to evaluate a particular definite integral. In your own words, summarize the process of evaluating a definite integral $\int_a^b f(x) dx$ using the definition of the integral. (Which step is hard to do by hand?)

2 Suppose $\int_0^5 f(t) dt = 10$. Evaluate four out of the five expressions below. (You do not have enough information to evaluate the remaining one—explain why!)

(a) $\int_0^5 7f(t) dt$

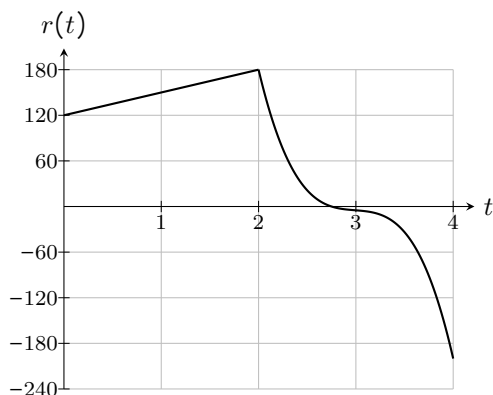
(b) $\int_0^5 (f(t) + 7) dt$

(c) $\int_0^5 f(t) dt + 7$

(d) $\int_0^5 7f(t + 7) dt$

(e) $\int_{-7}^{-2} 7f(t + 7) dt$

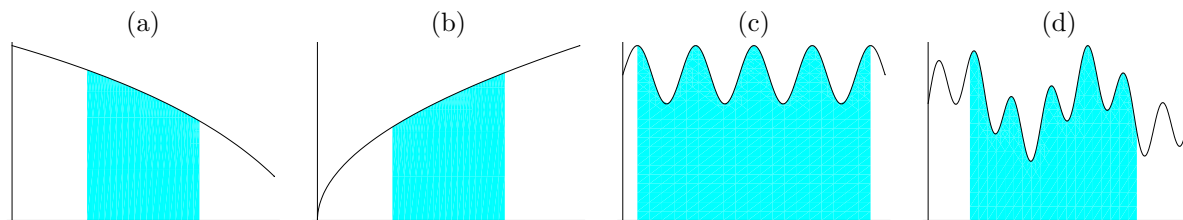
3 The function $r(t)$ graphed below gives the rate at which water is flowing into or out of a backyard swimming pool one afternoon. (Positive rates mean water is flowing into the pool.) Time t is measured in hours since noon, and the rate $r(t)$ is measured in gallons per hour. At 1 pm, there are 500 gallons of water in the pool.



- (a) Write an expression involving a definite integral that gives the exact amount of water in the pool at 3 pm.
- (b) Write an expression involving a definite integral that gives the exact amount of water in the pool at 2 pm, and then evaluate this expression exactly.
- (c) Is water flowing into or out of the pool between 1 pm and 2 pm? Based on that, should there be more water in the pool at 1 pm or at 2 pm? Make sure your answer to (b) is consistent with this. (Remember that we are told there are 500 gallons of water in the pool at 1 pm.)
- (d) Write an expression involving a definite integral that gives the exact amount of water in the pool x hours after noon.

4

Suppose we use right-hand and left-hand sums to estimate the areas pictured below. Match the pictures with the descriptions below, and briefly explain your choices.



Choices:

- A. L_4 will give an underestimate for this area, R_4 an overestimate.
- B. L_4 will give an overestimate, R_4 an underestimate.
- C. It is very hard to tell from the picture whether L_4 and R_4 will be overestimates or underestimates.
- D. Both L_4 and R_4 will give overestimates.